

Homework 4 Solution

1. Let $\mathcal{E} = (1, x, x^2)$ be the standard ordered basis of $(\mathbb{Z}_5)_2[x]$ and let $\mathcal{B} = \left(\vec{w}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \vec{w}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \vec{w}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}\right)$ be the ordered basis of the linear space given by

$$W = \left\{ \begin{pmatrix} w \\ x \\ y \\ z \end{pmatrix} \in \mathbb{Z}_5^4 : z = w + x + y \right\}$$

Note: you do not have to prove that W is a subspace of $\mathbb{Z}_5^4$ nor that $\mathcal{B}$ is an ordered basis of W .

- (a) Let $S : (\mathbb{Z}_5)_2[x] \rightarrow W$ be a linear transformation. Suppose that

$$[S]_{\mathcal{B}}^{\mathcal{E}} = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 4 & 4 \\ 2 & 1 & 3 \end{pmatrix}$$

Compute $S(a + bx + cx^2)$ for every $a + bx + cx^2 \in (\mathbb{Z}_5)_2[x]$.

Solution: Let $a + bx + cx^2 \in (\mathbb{Z}_5)_2[x]$. By the definition of the representation matrix,

$$S(1) = \vec{w}_1 + 2\vec{w}_3 = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}, \quad S(x) = 4\vec{w}_2 + \vec{w}_3 = \begin{pmatrix} 0 \\ 4 \\ 1 \end{pmatrix}, \quad S(x^2) = 2\vec{w}_1 + 4\vec{w}_2 + 3\vec{w}_3 = \begin{pmatrix} 2 \\ 4 \\ 3 \end{pmatrix}$$

It follows that

$$S(a + bx + cx^2) \stackrel{\text{LT}}{=} aS(1) + bS(x) + cS(x^2) = a \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + b \begin{pmatrix} 0 \\ 4 \\ 1 \end{pmatrix} + c \begin{pmatrix} 2 \\ 4 \\ 3 \end{pmatrix} = \begin{pmatrix} a + 2c \\ 4b + 4c \\ 2a + b + 3c \end{pmatrix}$$

- (b) Consider the linear transformation $T : (\mathbb{Z}_5)_2[x] \rightarrow W$ defined by

$$T(a + bx + cx^2) = \begin{pmatrix} a + 4b \\ c + b + 3a \\ 2a + c \\ a + 2c \end{pmatrix} \quad \forall a + bx + cx^2 \in (\mathbb{Z}_5)_2[x]$$

Compute the representation matrix $[T]_{\mathcal{B}}^{\mathcal{E}}$.

Solution: We compute

$$T(1) = \begin{pmatrix} 1 \\ 3 \\ 2 \\ 1 \end{pmatrix} = \vec{w}_1 + 3\vec{w}_2 + 2\vec{w}_3, \quad T(x) = \begin{pmatrix} 4 \\ 1 \\ 0 \\ 0 \end{pmatrix} = 4\vec{w}_1 + \vec{w}_2, \quad T(x^2) = \begin{pmatrix} 0 \\ 1 \\ 1 \\ 2 \end{pmatrix} = \vec{w}_2 + \vec{w}_3$$

It follows that

$$[T]_{\mathcal{B}}^{\mathcal{E}} = \begin{pmatrix} 1 & 4 & 0 \\ 3 & 1 & 1 \\ 2 & 0 & 1 \end{pmatrix}$$

- (c) Prove that T is invertible and compute the representation matrix $[T^{-1}]_{\mathcal{E}}^{\mathcal{B}}$.

Hint. Use the inversion algorithm.

Solution:

We know that $[T^{-1}]_{\mathcal{E}}^{\mathcal{B}} = ([T]_{\mathcal{B}}^{\mathcal{E}})^{-1}$, so we apply the inversion algorithm to $[T]_{\mathcal{B}}^{\mathcal{E}}$

$$\begin{aligned} \left([T]_{\mathcal{B}}^{\mathcal{E}} \mid I_3 \right) &= \left(\begin{array}{ccc|ccc} 1 & 4 & 0 & 1 & 0 & 0 \\ 3 & 1 & 1 & 0 & 1 & 0 \\ 2 & 0 & 1 & 0 & 0 & 1 \end{array} \right) \xrightarrow[R_3 \rightarrow R_3 + 3R_1]{R_2 \rightarrow R_2 + 2R_1} \left(\begin{array}{ccc|ccc} 1 & 4 & 0 & 1 & 0 & 0 \\ 0 & 4 & 1 & 2 & 1 & 0 \\ 0 & 2 & 1 & 3 & 0 & 1 \end{array} \right) \\ &\xrightarrow{R_3 \rightarrow R_3 + 2R_2} \left(\begin{array}{ccc|ccc} 1 & 4 & 0 & 1 & 0 & 0 \\ 0 & 4 & 1 & 2 & 1 & 0 \\ 0 & 0 & 3 & 2 & 2 & 1 \end{array} \right) \xrightarrow[R_3 \rightarrow 2R_3]{R_2 \rightarrow 4R_2} \left(\begin{array}{ccc|ccc} 1 & 4 & 0 & 1 & 0 & 0 \\ 0 & 1 & 4 & 3 & 4 & 0 \\ 0 & 0 & 1 & 4 & 4 & 2 \end{array} \right) \\ &\xrightarrow[R_2 \rightarrow R_2 + R_3]{R_1 \rightarrow R_1 + R_2} \left(\begin{array}{ccc|ccc} 1 & 0 & 4 & 4 & 4 & 0 \\ 0 & 1 & 0 & 2 & 3 & 2 \\ 0 & 0 & 1 & 4 & 4 & 2 \end{array} \right) \xrightarrow{R_1 \rightarrow R_1 + R_3} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 3 & 3 & 2 \\ 0 & 1 & 0 & 2 & 3 & 2 \\ 0 & 0 & 1 & 4 & 4 & 2 \end{array} \right) \end{aligned}$$

It follows that T is invertible and

$$[T^{-1}]_{\mathcal{E}}^{\mathcal{B}} = ([T]_{\mathcal{B}}^{\mathcal{E}})^{-1} = \begin{pmatrix} 3 & 3 & 2 \\ 2 & 3 & 2 \\ 4 & 4 & 2 \end{pmatrix}$$

2. Let V be a vector space over $\mathbb{R}$ with $\dim(V) = 3$. Let $\mathcal{A} = (v_1, v_2, v_3)$ and $\mathcal{B} = (w_1, w_2, w_3)$ be two ordered bases of V and let $v, w \in V$. Suppose that

$$[id_V]_{\mathcal{B}}^{\mathcal{A}} = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

- (a) Suppose that $[w]_{\mathcal{B}} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$. Compute $[w]_{\mathcal{A}}$.

Solution:

Since $[id_V]_{\mathcal{B}}^{\mathcal{A}}$ is upper triangular, $\det[id_V]_{\mathcal{B}}^{\mathcal{A}} = 1 \cdot 1 \cdot 1 = 1$. Using the adjugate matrix,

$$[id_V]_{\mathcal{A}}^{\mathcal{B}} = ([id_V]_{\mathcal{B}}^{\mathcal{A}})^{-1} = \frac{\text{adj}[id_V]_{\mathcal{B}}^{\mathcal{A}}}{\det[id_V]_{\mathcal{B}}^{\mathcal{A}}} \stackrel{\text{direct formula}}{=} \begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}.$$

Thus

$$[w]_{\mathcal{A}} = [id_V]_{\mathcal{A}}^{\mathcal{B}} \cdot [w]_{\mathcal{B}} = \begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix}$$

- (b) Suppose that $[v]_{\mathcal{B}} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$. Compute $[v + w]_{\mathcal{A}}$.

Solution: We know

$$[v]_{\mathcal{A}} = [id_V]_{\mathcal{A}}^{\mathcal{B}} \cdot [v]_{\mathcal{B}} = \begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}.$$

It follows that

$$[v + w]_{\mathcal{A}} = [v]_{\mathcal{A}} + [w]_{\mathcal{A}} = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} + \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} -2 \\ -1 \\ 2 \end{pmatrix}.$$

3. Let V be a vector space over $\mathbb{R}$ for which $\dim(V) = 2$. Let $\mathcal{A} = (v_1, v_2)$ and $\mathcal{B} = (w_1, w_2)$ be two ordered bases of V and let $T : V \rightarrow V$ be a linear operator. Suppose that

$$[T]_{\mathcal{A}}^{\mathcal{A}} = \begin{pmatrix} 1 & 2 \\ 3 & -1 \end{pmatrix} \quad \text{and} \quad [T]_{\mathcal{A}}^{\mathcal{B}} = \begin{pmatrix} 3 & 2 \\ -1 & 4 \end{pmatrix}$$

- (a) Compute the transition matrices $[id]_{\mathcal{A}}^{\mathcal{B}}$ and $[id]_{\mathcal{B}}^{\mathcal{A}}$.

Hint. We have $[T]_{\mathcal{A}}^{\mathcal{B}} = [T]_{\mathcal{A}}^{\mathcal{A}} \cdot [id]_{\mathcal{A}}^{\mathcal{B}}$.

Solution: We have

$$[T]_{\mathcal{A}}^{\mathcal{B}} = [T]_{\mathcal{A}}^{\mathcal{A}} \cdot [id]_{\mathcal{A}}^{\mathcal{B}}$$

It follows that

$$[id]_{\mathcal{A}}^{\mathcal{B}} = ([T]_{\mathcal{A}}^{\mathcal{A}})^{-1} \cdot [T]_{\mathcal{A}}^{\mathcal{B}} \stackrel{\text{formula}}{=} \frac{1}{7} \begin{pmatrix} 1 & 2 \\ 3 & -1 \end{pmatrix} \cdot \begin{pmatrix} 3 & 2 \\ -1 & 4 \end{pmatrix} = \frac{1}{7} \begin{pmatrix} 1 & 10 \\ 10 & 2 \end{pmatrix}$$

It follows that

$$[id]_{\mathcal{B}}^{\mathcal{A}} = ([id]_{\mathcal{A}}^{\mathcal{B}})^{-1} \stackrel{\text{formula}}{=} \frac{1}{14} \begin{pmatrix} -2 & 10 \\ 10 & -1 \end{pmatrix}$$

We can also compute $[id]_{\mathcal{B}}^{\mathcal{A}}$ in an alternative way:

$$[id]_{\mathcal{B}}^{\mathcal{A}} = [T^{-1} \circ T]_{\mathcal{B}}^{\mathcal{A}} = [T^{-1}]_{\mathcal{B}}^{\mathcal{A}} \cdot [T]_{\mathcal{A}}^{\mathcal{A}} = ([T]_{\mathcal{A}}^{\mathcal{B}})^{-1} \cdot [T]_{\mathcal{A}}^{\mathcal{A}} \stackrel{\text{formula}}{=} \frac{1}{14} \begin{pmatrix} 4 & -2 \\ 1 & 3 \end{pmatrix} \cdot \begin{pmatrix} 1 & 2 \\ 3 & -1 \end{pmatrix} = \frac{1}{14} \begin{pmatrix} -2 & 10 \\ 10 & -1 \end{pmatrix}$$

- (b) Compute $[T]_{\mathcal{B}}^{\mathcal{B}}$.

Solution: We have

$$\begin{aligned} [T]_{\mathcal{B}}^{\mathcal{B}} &= [id \circ T \circ id]_{\mathcal{B}}^{\mathcal{B}} = [id]_{\mathcal{B}}^{\mathcal{A}} [T]_{\mathcal{A}}^{\mathcal{A}} [id]_{\mathcal{A}}^{\mathcal{B}} = \frac{1}{14} \begin{pmatrix} -2 & 10 \\ 10 & -1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 2 \\ 3 & -1 \end{pmatrix} \cdot \frac{1}{7} \begin{pmatrix} 1 & 10 \\ 10 & 2 \end{pmatrix} \\ &= \frac{1}{14 \cdot 7} \begin{pmatrix} 28 & -14 \\ 7 & 21 \end{pmatrix} \cdot \begin{pmatrix} 1 & 10 \\ 10 & 2 \end{pmatrix} = \frac{1}{14 \cdot 7} \begin{pmatrix} -112 & 252 \\ 217 & 112 \end{pmatrix} = \frac{1}{14} \begin{pmatrix} -16 & 36 \\ 31 & 16 \end{pmatrix} \end{aligned}$$

Alternative solution: We can also write

$$[T]_{\mathcal{B}}^{\mathcal{B}} = [id \circ T]_{\mathcal{B}}^{\mathcal{B}} = [id]_{\mathcal{B}}^{\mathcal{A}} \cdot [T]_{\mathcal{A}}^{\mathcal{B}} = \frac{1}{14} \begin{pmatrix} -2 & 10 \\ 10 & -1 \end{pmatrix} \cdot \begin{pmatrix} 3 & 2 \\ -1 & 4 \end{pmatrix} = \frac{1}{14} \begin{pmatrix} -16 & 36 \\ 31 & 16 \end{pmatrix}$$

4. Let V be a finite-dimensional vector space over the field F and let $T, S \in \text{hom}(V, V)$. Let $\mathcal{A}$ and $\mathcal{B}$ be two ordered bases of V . Suppose

that the following matrix identity holds:

$$[S]_{\mathcal{A}}^{\mathcal{B}} \cdot [T]_{\mathcal{B}}^{\mathcal{A}} = [T]_{\mathcal{A}}^{\mathcal{B}} \cdot [S]_{\mathcal{B}}^{\mathcal{A}}.$$

Prove that $S \circ T = T \circ S$.

Solution: We know

$$[S]_{\mathcal{A}}^{\mathcal{A}} = [id_V]_{\mathcal{A}}^{\mathcal{B}} \cdot [S]_{\mathcal{B}}^{\mathcal{B}} \cdot [id_V]_{\mathcal{B}}^{\mathcal{A}}$$

Using the hypothesis, then

$$[S \circ T]_{\mathcal{A}}^{\mathcal{A}} = [S]_{\mathcal{A}}^{\mathcal{B}} \cdot [T]_{\mathcal{B}}^{\mathcal{A}} = [T]_{\mathcal{A}}^{\mathcal{B}} \cdot [S]_{\mathcal{B}}^{\mathcal{A}} = [T \circ S]_{\mathcal{A}}^{\mathcal{A}}$$

Since two representation matrices are equal with respect to the same basis, then the corresponding linear transformations are equal. Thus, $S \circ T = T \circ S$.

5. Let V be a vector space over a field F and let $\mathcal{A} = (v_1, \dots, v_n)$ be an ordered basis of V . Let $T : V \rightarrow V$ be a linear operator. Suppose that $\mathcal{B} = (T(v_1), \dots, T(v_n))$ is an ordered basis of V . Prove that

$$[T]_{\mathcal{A}}^{\mathcal{A}} = [T]_{\mathcal{B}}^{\mathcal{B}}$$

Solution:

First, note that by definition, for every $1 \leq j \leq n$ we have

$$[T(v_j)]_{\mathcal{B}} = \vec{e}_j$$

It follows that $[T]_{\mathcal{B}}^{\mathcal{A}} = I_n$. Thus

$$[T]_{\mathcal{A}}^{\mathcal{A}} = [id_V \circ T]_{\mathcal{A}}^{\mathcal{A}} = [id_V]_{\mathcal{A}}^{\mathcal{B}} [T]_{\mathcal{B}}^{\mathcal{A}} = [id_V]_{\mathcal{A}}^{\mathcal{B}} = [T]_{\mathcal{B}}^{\mathcal{A}} \cdot [id_V]_{\mathcal{A}}^{\mathcal{B}} = [T \circ id_V]_{\mathcal{B}}^{\mathcal{B}} = [T]_{\mathcal{B}}^{\mathcal{B}}$$

6. Prove or disprove each of the following statements:

- (a) The matrices given by

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} 1 & 0 \\ 1 & 2 \end{pmatrix}$$

are similar.

Solution:

True. We search for an invertible matrix $P = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_2(\mathbb{R})$ such that $AP = PB$, namely

$$\begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = AP = PB = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 2 \end{pmatrix}$$

Equivalently,

$$\begin{pmatrix} a+c & b+d \\ 2c & 2d \end{pmatrix} = \begin{pmatrix} a+b & 2b \\ c+d & 2d \end{pmatrix}$$

which leads to the following homogeneous linear system in the variables a, b, c, d

$$\begin{cases} a + c = a + b \\ b + d = 2b \\ 2c = c + d \\ 2d = 2d \end{cases} \Rightarrow c = b, b = d, c = d$$

We choose $a = 0, b = c = d = 1$ and consequently $P = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$. Note that $\det P = -1 \neq 0$ and therefore P is invertible. Note also that $AP = PB$. Therefore, A, B are similar.

(b) The matrices given by

$$A = \begin{pmatrix} 1+i & 2 \\ i & 1-2i \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} 1+i & 2 \\ 2i & 1-i \end{pmatrix}$$

are similar.

Solution:

False. A, B are not similar, as $\text{trace}(A) \neq \text{trace}(B)$. Indeed,

$$\text{trace}(A) = (1+i) + (1-2i) = 2-i \quad \text{trace}(B) = (1+i) + (1-i) = 2$$

Alternative solution:

A, B are not similar as $\det(A) \neq \det(B)$. Indeed

$$\det(A) = (1+i)(1-2i) - 2i = 1 - 2i + i + 2 - 2i = 3 - 3i$$

whereas

$$\det(B) = (1+i)(1-i) - 4i = 2 - 4i$$

(c) Let $K = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. Then the set

$$W = \{A \in M_2(\mathbb{R}) : A, K \text{ are not similar}\}$$

is a subspace of $M_2(\mathbb{R})$.

Solution:

False. We prove that W is not closed under addition.

We choose $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. Note that

$$\text{trace}(A) = 2 \neq 0 = \text{trace}(K)$$

and therefore A, K are not similar, which implies that $A \in W$.

We choose $B = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$. Note that

$$\text{trace}(B) = -2 \neq 0 = \text{trace}(K)$$

and therefore B, K are not similar, which implies that $B \in W$. On the other hand $A + B = K$, and as every matrix is always similar to itself, then $A + B \notin W$. We conclude that W is not closed under addition, and therefore W is not a subspace of $M_2(\mathbb{R})$.

(d) Let $\alpha, \beta \neq 0$. Then the matrices $A = \begin{pmatrix} 0 & \alpha \\ 0 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 0 \\ \beta & 0 \end{pmatrix}$ are similar.

Solution:

True. We search for an invertible matrix $P = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_2(\mathbb{R})$ for which $AP = PB$, namely

$$\begin{pmatrix} 0 & \alpha \\ 0 & 0 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = AP = PB = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 0 & 0 \\ \beta & 0 \end{pmatrix}$$

Developing both sides of the equality gives

$$\begin{pmatrix} \alpha c & \alpha d \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} \beta b & 0 \\ \beta d & 0 \end{pmatrix}$$

which leads to the following linear system in the variables a, b, c, d

$$\begin{cases} \alpha c & = & \beta b \\ \alpha d & = & 0 \\ \beta d & = & 0 \end{cases} \xRightarrow{\alpha, \beta \neq 0} d = 0, c = \frac{\beta}{\alpha} b$$

Choosing $a = d = 0, b = \alpha$ and $c = \beta$ guarantees a solution for the above system. The matrix P is given by $P = \begin{pmatrix} 0 & \alpha \\ \beta & 0 \end{pmatrix}$ which

clearly is invertible (two linearly independent rows). We conclude that A, B are similar and that $P^{-1}AP = B$ with P given by

$$P = \begin{pmatrix} 0 & \alpha \\ \beta & 0 \end{pmatrix}.$$